

Medius Bridge

Getting started with your hardware and Vision Labs

A simple end-user guide

Use this guide to connect a new Medius box, install its firmware, and configure the Medius Bridge plugin. No programming knowledge is required.

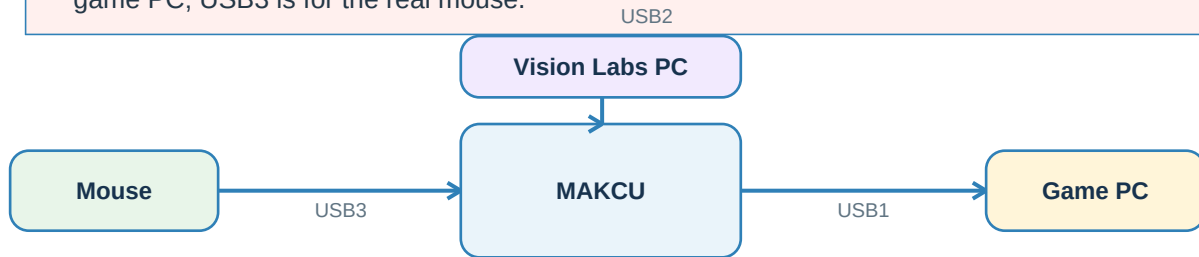
Before you begin: Have the Medius firmware folder, three USB cables, your mouse, a Vision Labs PC, and a separate game PC ready.

1. Connect the hardware

Start with only the control cable connected. This makes the first firmware installation easier.

Port	Connect it to	What it does
USB1	Game PC	Sends the mouse signal to the game
USB2	Vision Labs PC	Controls the box and runs the plugin
USB3	Your real mouse	Receives your physical mouse input

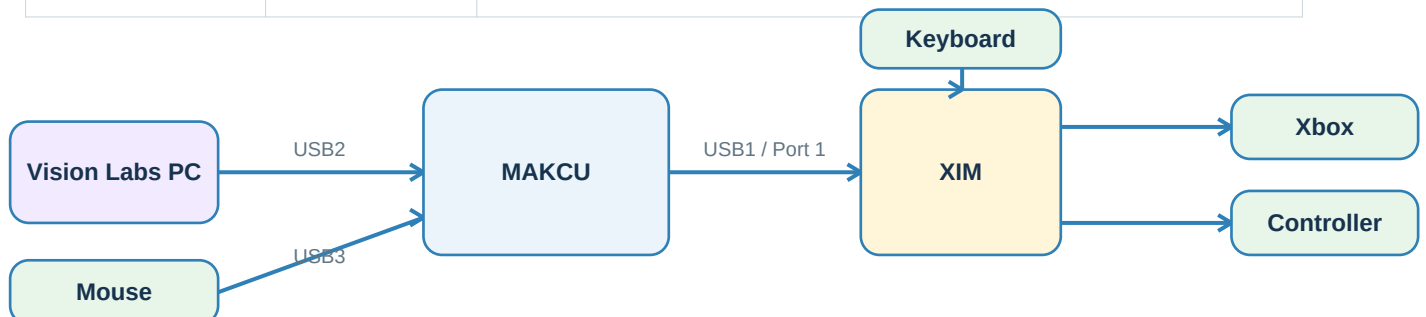
Important safety rule: Never connect USB1 and USB3 to the same computer. USB1 goes to the game PC; USB3 is for the real mouse.



Console setup using a XIM

If you are using a console through a XIM, use this exact wiring:

Device	Port	Connect to
MAKCU	USB 1	XIM
MAKCU	USB 2	Vision Labs PC
MAKCU	USB 3	Mouse
XIM	Port 1	MAKCU
XIM	Port 2	Keyboard
XIM	Port 3	Xbox controller
XIM	Port 4	Xbox



For a console setup, the **Vision Labs PC connects to MAKCU USB 2**. The XIM connects to MAKCU USB 1, and the mouse connects to MAKCU USB 3. Keep the Xbox and controller connected to the XIM as shown above.

2. Install the firmware

If your box already has working Medius firmware, skip to the next page. For a new box, use the factory flash files supplied with the hardware.

1. Install the CH343/CH34x driver if Windows does not recognize the box.
2. Install Espressif **esptool**.
3. Open the Medius firmware folder.
4. Run the included **factory flash** script.
5. When asked, select the CH343 COM port shown in Windows Device Manager.
6. When the script finishes, unplug and reconnect the box.

If the flash script cannot find the box, hold the board's **BOOT** button while reconnecting or powering it, then run the script again. Windows may assign a new COM number.

3. Install and configure the plugin

Once the firmware is installed, complete the setup on the Vision Labs PC.

1. Copy the **MediusBridge** folder into **VisionLabs\Plugins**, or install the supplied **Medius_Bridge.vlplugin** file.
2. Restart Vision Labs.
3. Open **Plugins**, enable **Medius Bridge**, and approve the permission review.
4. Leave **Serial port** blank so the plugin can find the box automatically.
5. Wait for the plugin log to show a message similar to connected fw 3.3.4 proto 6.

4. Set the Vision Labs mouse output

In Vision Labs settings, set the mouse output backend to:

Setting	Value
Mouse output backend	KMBoxNet
IP address	127.0.0.1
Port	8808
UUID	Any value, for example ABCD12345

In the Medius Bridge settings, keep **Output source** set to **Vision Labs native tracking (KMBoxNet emulation)**. This is the recommended mode.

5. Test the complete setup

With the plugin connected, connect the remaining cables:

USB3 to the real mouse **USB1** to the game PC **USB2** to the Vision Labs PC

Before opening a game, check the Vision Labs plugin log. You should see that the Medius box is connected and that KMBBoxNet traffic is arriving. The game PC should detect the box as a mouse.

First test: Move the physical mouse. It should still work normally. Then enable your Vision Labs profile and confirm that the plugin can send movement through the box.

Common problems

What you see	What to check
Plugin says no box found	USB2 is connected to the Vision Labs PC, the CH343 driver is installed, and no other program is using the COM port.
No mouse on the game PC	USB1 is connected to the game PC and USB3 is connected to the mouse. Never put both on one computer.
Flash script cannot find hardware	Hold BOOT while reconnecting the box, check the new COM number, and run the factory script again.
Plugin connects but no movement	Vision Labs is set to KMBBoxNet, IP 127.0.0.1, port 8808, and the plugin output source is the KMBBoxNet emulation mode.

If the plugin stops or loses connection, the box automatically returns to normal mouse passthrough after about one second. This prevents a stuck injected input.

You are ready to use Medius Bridge with Vision Labs.

This guide intentionally covers only the end-user setup. Firmware protocol details, API references, and developer notes are kept in the project source and Medius technical documentation.